

ZYNVEX SOFTWARE ENGINEERING INTERNSHIP

MODULE 4 TECHNICAL REPORT: PORTALS & ADMIN CONTROL

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Project Name: QuickServe

Submission Date: August 21, 2026

1. Module Overview

Module 4 finalized the staff portals. This included developing the Kitchen Display System (KDS) featuring Kanban order tracking columns, building the admin menu manager with full database CRUD functionalities, and creating a feedback manager for reviews and complaints.

2. Core Deliverables Met

- Kitchen Kanban Board:** Designed the `/kitchen`` view with three statuses: Pending, Preparing, and Ready. Orders move across columns in real-time.
- Kitchen ETA Broadcast:** Integrated action buttons inside the KDS. Transitioning an order to "Preparing" automatically calculates a +15 minute completion timestamp and writes it back to Supabase.
- Admin Analytics Panel:** Built dynamic dashboards calculating key business indicators (Total Revenue, Active Menu Items, Pending Orders).
- Full CRUD Menu Management:** Built `/admin/menu`` enabling management of menu entries (Add, Edit, Toggle availability, Delete) directly from the browser.
- Reviews & Complaints Dashboard:** Built `/admin/reviews`` to display customer feedback in real-time, automatically flagging complaints for manager attention.

3. Key Files & Architecture

- `src/app/kitchen/page.tsx``: Kitchen display system with drag-free column transitions and status updates.
- `src/app/admin/page.tsx``: Owner analytics landing page.
- `src/app/admin/menu/page.tsx``: CRUD interface for item catalogs.
- `src/app/admin/reviews/page.tsx``: Review dashboard with complaint flags.
- `src/app/admin/layout.tsx``: Layout with permanent side navigation and secure sign-out.

4. Verification & Testing

- Kitchen to Customer Pipeline:** Successfully verified that marking an order "Preparing" in `/kitchen`` immediately writes the ETA and updates the customer `/order/[id]`` countdown.
- CRUD Validation:** Adding or modifying menu items inside the admin dashboard shows instant updates on the customer `/menu`` catalog.